Ahmed Ikram | Software Engineer

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Final-year BSc Computer Science (Data Science & AI), University of Dundee (predicted First, June 2027). Shipped full-lifecycle software - design to deploy to real-user evaluation, fixing the issues users surfaced before deployment. Learned Terraform and cloud networking from zero inside an industry project where I was Team Lead for a 7-person Agile team. Worked in standard PR and code-review conventions in university group project teams. Seeking a graduate Software Engineering role from June 2027.

EDUCATION

University of Dundee - *BSc (Hons) Computer Science (Data Science and AI)* *Sep 2023 - Present*

- On track for a First-Class Honours degree. Expected Graduation: June 2027
- Honours Project: Tiny Language Model Laboratory - Configurable Transformer Training Framework

  Engineering a modular research framework for configuring, training and evaluating miniature Transformer language models. Implementing reusable components for synthetic task generation, tokenisation, model construction, training loops, checkpointing, reproducibility and experiment visualisation, with the system designed to support repeatable parameterised experiments rather than a single fixed model.

High School of Dundee *Sep 2017 - June 2023*

- Highers: A Mathematics, A Computing Science, A English, B Chemistry. Advanced Highers: Computing Science C.
- Computing Science classroom assistant - mentored junior pupils one-to-one during classwork.

TECHNICAL SKILLS

Languages: Python · TypeScript · JavaScript · SQL · T-SQL · Rust / PyO3 · Java · C · Bash

Backend, APIs, Databases: FastAPI · Flask · asyncio · GraphQL (Strawberry) · Pydantic · SQLAlchemy · Apache Kafka · Model Context Protocol (MCP) · PostgreSQL · SQLite · Redis · Alembic

Frontend: React · React Native · Expo · Tailwind CSS · Vite · Formik · GraphQL Codegen · graphql-ws · Socket.IO

Cloud & DevOps: AWS (ECS Fargate, EC2, S3, RDS, ElastiCache, WAF, IAM, CloudFront, Secrets Manager, DynamoDB) · GCP (GKE, Cloud SQL, Cloud Monitoring) · Kubernetes (Kustomize, HPA, PDB, NetworkPolicy) · Azure (Databricks, Azure SQL, ADLS) · Terraform · Docker · Linux · GitHub Actions

Observability & Testing: Grafana · Prometheus · OpenTelemetry (Jaeger) · StatsD · Pytest · Jest · Playwright · Cypress · k6

PROJECTS

DevSync Project Management Platform with GitHub Integration

React · Tailwind · Flask · PostgreSQL · Socket.IO · GitHub OAuth · GCP · GKE · Terraform · GitHub Actions CI/CD

- Realtime boards (tasks, live comments, presence) on JWT Socket.IO rooms that re-verify project membership per-event and degrade gracefully instead of crashing; Redis message queue fans out across replicas with cross-pod receipt tests.
- Hardened GKE artifact security using GCP Workload Identity, Cosign image signing, Trivy scanning, and Binary Authorization to enforce SLSA L3 provenance.
- Engineered a highly available GKE platform integrated with automated CD pipelines, featuring horizontal autoscaling, Cloud Armor WAF protection, and automated e2e/load testing on merge.
- GitHub OAuth integration: issue/PR linking and comment write-back with encrypted token storage; 3-tier RBAC enforced server-side.
- 1,634 tests and CodeQL static analysis gate every merge, p95 100ms vs a 500ms ceiling at 250/250 passing checks; a google managed Prometheus dashboard flags backend downtime.

ATM Log Aggregation, Anomaly Detection & Diagnostics (NCR Atleos)

FastAPI · React · PostgreSQL · Kafka · Redis · Docker · Terraform · AWS · CI/CD

- Architected a FastAPI REST API (JWT & Redis-blacklist auth) with health-check-cascading startup, with a zero-dependency typed decision-API adapter with per-error-class retries and circuit breaker with graceful fallback.
- Built the Kafka (KRaft) event pipeline: at-least-once delivery, hybrid Redis/LRU deduplication, Redis Stream DLQ, sustaining ~100 msg/s over 2.5M events; W3C trace context via Kafka headers end-to-end, with OpenTelemetry Collector exporting traces to Jaeger.

- Applied 8 Redis patterns (rate-limit, dedup, lock, pub/sub, cache, JWT blacklist, DLQ, counters) off one connection pool; React frontend ships as a ~25MB nginx-alpine image measured at build time.
- Provisioned Terraform across a multi-AZ VPC (ECS Fargate behind ALB, EC2-hosted Kafka & Redis in a private subnet); a 3-pipeline CI/CD runs 1,974 tests (1,439 pytest, 495 Vitest, 10 Playwright E2E, 30 Terraform assertions) and Checkov compliance on every PR, rolling out zero-downtime with zero long-lived credentials. State locked in DynamoDB.

StockLens FinTech Spending & Investment Mobile App

React Native · Expo · FastAPI · PostgreSQL · Redis · Rust · GraphQL · MCP · Terraform · Docker · AWS · CI/CD

- Engineered an async FastAPI backend: 70+ REST endpoints with JWT auth (bcrypt), biometric unlock, and a stateless MCP server exposing 16 tools over streamable HTTP with OAuth 2.1 PKCE and JWT/JWKS validation.
- Replaced a pandas bottleneck with a native Rust extension (PyO3/Maturin, 12 functions) for technical-indicator computation, cross-built for ARM64 via QEMU in CI and shared by both the LSTM training pipeline and live API.
- Added a GraphQL read façade over the same service layer as REST: Strawberry resolvers with batched preloads eliminate N+1 queries, backed by a shared 60-second Redis quote cache; an authenticated graphql-ws subscription replaces client polling, and agent responses stream over an SSE endpoint via a custom fetch+ReadableStream parser.
- Deployed via Terraform: three-tier network isolation (WAF-protected ALB → ECS Fargate ARM64 → private RDS/Redis) for a 20-30% compute-cost cut over x86; multi-stage images are 450MB vs 1.2GB naive.
- 2,417 assertions - and a CodeQL static analysis - within a 9-job CI and an 8-job, 5-stage OIDC-authenticated CD pipeline.

W3C Web Logs ETL Pipeline

Airflow · Databricks · dbt · PySpark · Azure SQL · Terraform · Power BI · Grafana · Prometheus · Marquez · OpenLineage

- Architected a Databricks DLT medallion in PySpark (Bronze → Silver in DLT; gold via dbt) on serverless distributed compute; 153,380 rows ingested with 3 Silver rows dropped through 10 DLT quality gates (7 bronze & 3 Silver).
- 16 dbt models compile against PostgreSQL and T-SQL from one codebase via 18 compatibility macros; 121 dbt tests gate every model, catching 213 duplicate rows through a 12-component deduplication key.
- Integrated OpenLineage with a self-hosted Marquez collector across 2 Airflow DAGs - task-run events, dbt model/column lineage, Azure SQL dataset edges, and custom data-quality facets in one cross-engine lineage graph.
- Deployed on Azure via Terraform and OIDC federation; 750 tests, a freshness probe, 3 Grafana dashboards (23 panels), 8 Prometheus rules and 2 Azure Monitor alerts validate pipeline health and data quality.

CERTIFICATES

AWS Academy Graduate - Machine Learning Foundations	F*eb 2025*
Foundations of Azure AI: Concepts, Capabilities, and Implementation	*Oct 2024*
AWS Academy Graduate - Cloud Foundations	*Mar 2024*

EXPERIENCE

Beauly Mini Market, Dundee, Scotland

Retail Assistant — *Sep 2023 - Present*

Zulus Peri Peri Grill, Dundee, Scotland

Front of House Worker — *June 2021 - Sep 2023*